

THE LATHE — COMPLETE REFERENCE POSTER

The “mother of machine tools”: a rotating workpiece + a moving single-point cutting tool = every cylindrical, conical, threaded & spherical part in the machine world

FIRST LATHE
~1300 BCE

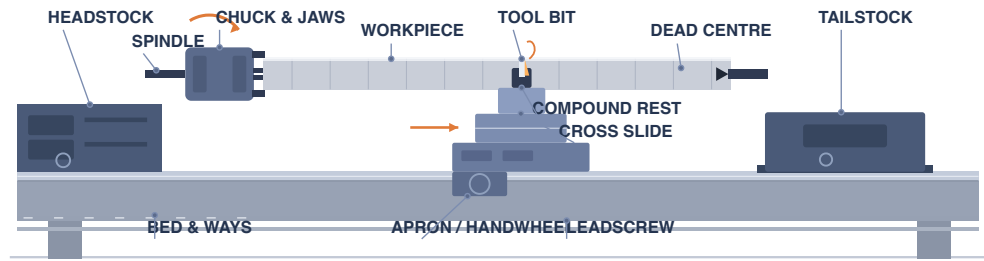
MAUDELAY 1797
SCREW CUTTING

MANUAL ACCURACY
±0.01 mm

CNC ACCURACY
±0.001 mm

1 ANATOMY OF AN ENGINE LATHE (CENTRE LATHE)

A **lathe** holds the workpiece and **spins it about its own axis** while a **single-point cutting tool** is pressed against it. Because the work rotates, every feature is **symmetrical about the axis** — cylinders, cones, threads, spheres, bores. It is the “**mother of machine tools**”: it can machine the parts to build every other machine tool, including itself.



■ Cast-iron members ■ Workpiece ■ Moving / driven parts ■ Cutting action & motion ■ HSS tool bit

Workpiece **ROTATES** about its axis (orange arrow); the tool is fed along the bed (longitudinal) or across it (*facing*) to remove material. The leadscrew synchronises carriage travel with spindle rotation so screw threads cut perfectly.

2 MAIN PARTS & THEIR JOBS

PART	FUNCTION
Bed	Heavy cast-iron base; keeps headstock, carriage & tailstock aligned; absorbs vibration.
Ways	Hardened ground guide rails the carriage slides along.
Headstock	Houses spindle, bearings, belts/gears and speed controls.
Spindle	Hollow rotating shaft driving the chuck/collet; bore lets long bar stock pass through.
Chuck	3-jaw self-centring (round stock) or 4-jaw independent (irregular/square work).
Carriage	Saddle sliding along the bed; carries cross slide & apron; longitudinal feed.
Cross slide	Moves the tool at 90° to the axis — facing and depth of cut.
Compound rest	Swivels to any angle on the cross slide — taper turning; holds the tool post.
Tool post	Clamps the tool bit or quick-change holder at centre height.
Apron	Front of carriage: feed gears, clutch and carriage handwheel.
Tailstock	Supports long work with a centre; quill (Morse taper) holds drills & reamers.
Leadscrew	Precision threaded shaft driving the carriage for screw cutting; geared 1:1 with spindle.
Feed rod	Power feed (turning/facing) at selected rates, no thread sync.
Centres	Dead (fixed) & live (rotating) 60° centres — work <i>between centres</i> for max accuracy.
Faceplate	Slotted plate for bolting on irregular work that can't be chucked.
Change gears	Set the leadscrew ratio to cut any thread pitch.

3 LATHE OPERATIONS — WHAT YOU CAN CUT

Facing Square the end of the work; tool feeds across the end face. First operation after chucking.	Straight turning Reduce diameter along the length — roughing (deep cuts) then finishing (light cuts, fine feed, accurate size).	Taper turning Cones by swivelled compound, offset tailstock, taper attachment or CNC — e.g. Morse tapers 1.5°–3°.
Grooving / parting Narrow tool cuts a groove (O-ring seats) or parts off the piece; thin tools — slow speed, firm workholding.	Knurling Hardened wheels press a diamond/straight grip pattern into handles — light pressure, slow speed, oil.	Thread cutting Single-point tool + leadscrew sync cuts 60° V-threads; dies cut external, taps cut internal threads.
Drilling Tailstock quill feeds the drill into rotating work — centre-drill first so it can't wander.	Boring Single-point boring bar enlarges/straightens holes; corrects holes drilled off-centre.	Reaming Finish a drilled hole to exact size and smoothness — removes only ~0.1–0.3 mm.
Eccentric turning Work between two offset centres makes crank pins and cam lobes.	Filing / polishing Slow speed, draw-file or emery cloth to smooth and finish the surface.	Hard turning Turn hardened steel (45–65 HRC) with CBN inserts — replaces grinding on many jobs.

Also possible: chamfering & deburring, spherical turning (form tool / CNC arc), tapping (internal threads via tailstock), and form-tool profile cutting.

4 WORKHOLDING, MEASURING & TOOLING

Workholding

- › **3-jaw scroll chuck** — self-centring, fastest; runout ~0.05–0.1 mm.
- › **4-jaw independent** — each jaw dialed in; squares & irregular; true to 0.01 mm.
- › **Collet / ER chuck** — precise ±0.005 mm; thin & bar stock, no jaw marks.
- › **Between centres** — dead/live centre + dog; most accurate for long shafts.
- › **Faceplate** — straps & bolts hold irregular castings.
- › **Steady / follower rests** — support long slender work; stop chatter.
- › **Mandrel** — pressed into the bore so the outside turns true to it.

Measuring

- › Vernier caliper 0.02 mm · micrometer 0.01 mm · dial indicator 0.01 mm · DRO 0.005 mm.
- › Check diameter at 3 positions, runout & taper — spindle **stopped**.

Tool materials

- › **HSS** — tough, cheap, grindable; ~600 °C max; the manual-lathe default. **Cobalt** for tough alloys.
- › **Carbide** — brazed or indexable inserts; 3–4× HSS speeds; rigid machines only.
- › **Coated carbide** (TiN/TiAlN) — less friction, longer life.
- › **Ceramic** — cast iron & hardened steel at very high speeds.
- › **CBN** — hard turning 45–65 HRC. **PCD** — aluminium & composites.

Insert code — CNMG 12 04 08

Pos	Meaning
C	Shape: C=80° rhombus, D=55°, T=triangle, S=square, V=35°
N	Relief angle: N=0°, C=7°, B=5° · M = tolerance
G	Hole + chipbreaker · 12 = size (mm) · 04 thick · 08 = 0.8 mm nose radius

Tool geometry

- › **Rake**: positive = sharp light cuts (HSS); negative = strong edge (carbide, high speed). **Clearance** ~5–10° stops flank rubbing.
- › Tip exactly at **spindle centre height**; bigger **nose radius** = stronger, smoother, higher forces.

5 SPEEDS, FEEDS & DEPTH OF CUT

Cutting speed (CS) is the workpiece surface speed in **m/min** — set by material, tool and operation, not the dial:

RPM = (CS × 1000) ÷ (π × D) D = diameter in mm · π ≈ 3.1416 · CS = π × D × RPM ÷ 1000 · Feed rate (mm/min) = feed (mm/rev) × RPM
Example: Ø40 mm mild steel, HSS, CS ≈ 35 m/min → RPM = 35 000 ÷ 125.7 ≈ 278 RPM ; carbide (CS 140) → ≈ 1114 RPM. Smaller Ø = higher RPM; bigger Ø = lower.

Workpiece material	HSS CS	Carbide CS	Feed fin./rough (mm/rev)
Mild / low-carbon steel	30–45	90–180	0.05–0.15 / 0.2–0.5
Aluminium alloys	60–120	200–400	0.05–0.15 / 0.15–0.4
Brass & bronze	45–75	150–250	0.05–0.15 / 0.15–0.4
Cast iron	20–30	60–120	0.05–0.15 / 0.2–0.5
Stainless steel	15–25	60–120	0.05–0.12 / 0.15–0.35
Plastics	60–120	200–300	0.05–0.1 / 0.1–0.3
Wood (hand chisels)	300–900 (hand feed)		by eye — light sweeping cuts

Rules of thumb

- › **Depth of cut:** roughing 1–4 mm (HSS) / 2–8 mm (carbide); finishing 0.1–0.5 mm. **Chatter** = too fast/deep, dull tool, too much overhang — slow down.
- › **Threading:** several light passes, finish with a zero-cut spring pass. **Coolant:** soluble oil (steel), paraffin (aluminium), dry (cast iron).

7 SAFETY & DAILY CARE

Before you touch the spindle

- › **No gloves, long sleeves, loose hair or jewellery** near a rotating chuck — they wrap and pull you in.
- › **Remove the chuck key** before starting — a thrown key is a lethal missile.
- › **Stop the spindle** before measuring, adjusting, cleaning or changing tools.
- › **Safety glasses always**; clear swarf with a brush/hook, never bare hands.
- › Secure work firmly; right speed; listen for chatter; never grab the chuck to stop it — use the brake. Guards closed, ventilated; know the **E-stop**.

Daily care

- › Brush chips after every job; **wipe the ways** — grit grinds the slides.
- › **Oil** ways, leadscrew & gears daily; a thin film stops rust.
- › Check & adjust **gibs** (no backlash) and belt tension; clean chuck jaws, re-check runout.
- › Align tailstock to headstock (test between centres); check tool tip height; keep electrical box dry.

6 TYPES OF LATHES

- › **Engine / centre lathe** — classic general-purpose manual lathe; belts & gears; turning, facing, threading.
- › **Bench / mini lathe** — small; models, jewellery, electronics, hobby work.
- › **Toolroom lathe** — highest manual precision; fine leadscrew, close tolerances; tool & die work.
- › **Gap-bed lathe** — removable bed section swings very large diameters past the carriage.
- › **Turret / capstan lathe** — multi-tool turret for repeat production batches.
- › **CNC turning centre** — computer control; canned threading, live tooling, C-axis milling, bar feeders.
- › **Swiss-type lathe** — sliding headstock + guide bushing; ultra-small precise parts (watch screws, medical).
- › **Vertical lathe (VTL)** — vertical spindle for heavy, large-diameter discs & flywheels.
- › **Wood lathe** — hand-held gouges & chisels on a tool rest; work turns, tool guided by hand.
- › **Metal-spinning lathe** — presses sheet metal discs over a rotating former.

Lathe vs mill: a lathe rotates the work past a fixed tool (circles about the axis: shafts, threads, bores); a mill rotates the tool and moves the work (flats, slots, 3D shapes). Turning centres add milling via live tooling; turn-mill centres do both in one setup.

8 WHY IT MATTERS

Every engine, gearbox, axle, bolt, bearing and screw on Earth is shaped by a lathe or by machines the lathe built. Without Maudslay's screw-cutting lathe (1797) there would be no standardised threads, no interchangeable parts, no mass production. Today's CNC turning centres hold ±0.001 mm and run unattended overnight — yet the **manual engine lathe is still the first machine every machinist learns on**, because it shows directly how metal behaves under a cutting edge.

Facts

- › Wood lathes hold the tool by hand; metal lathes clamp it rigidly — same motion, different rules.
- › Watch screws from Swiss lathes are thinner than a human hair; the largest lathes turn ship propeller shafts many metres long.
- › A lathe is the only machine tool that can build a copy of itself.

QUICK FACTS

- › “**Mother of machine tools**” — it makes the parts for every other machine tool.
- › **Maudslay 1797** — standardised threads still fit worldwide today.
- › **Biggest lathes** turn ship shafts & rotor discs many metres across.

EVOLUTION OF THE LATHE — 3300 YEARS

~**1300 BCE** Egypt: bow-driven wood lathes → ~**500 BCE** Roman two-person lathes → **Middle Ages** pole lathes (treadle + springy branch) → ~**1480** Da Vinci's flywheel treadle lathe → **1568** Besson: first screw cutting → **1797** Maudslay: all-metal screw-cutting lathe, slide rest, standard threads → **1840s–60s** turret lathes → **1890s** geared-headstock engine lathe → **1950s** punched-tape NC → **1970s** → microprocessor CNC turning centres.